

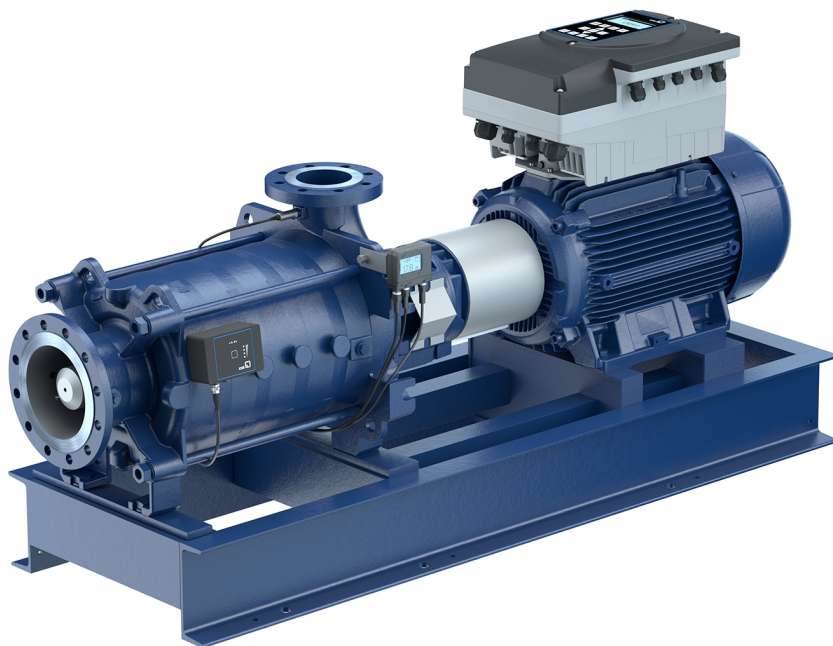


Centrifugal Pump

MultiTec Plus

High-pressure Ring-section Pump

Type Series Booklet



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Type Series Booklet MultiTec Plus

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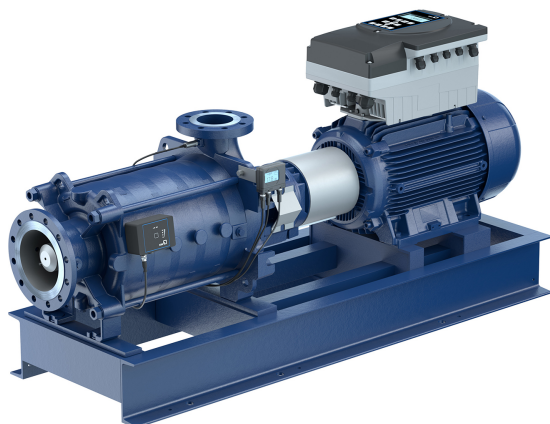
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Centrifugal Pumps

Multistage High-pressure Centrifugal Pumps

MultiTec Plus



Main applications

- Water supply
- Drinking water supply

Fluids handled

- Water¹⁾
- Drinking water¹⁾

Operating data

Table 1: Operating properties

Characteristic		Value
Size	DN	125 - 150
Flow rate	Q [m ³ /h]	≤ 470
	Q [l/s]	≤ 130,5
Head	H [m]	≤ 250
Fluid temperature	T [°C]	≥ 0
		≤ +60
Operating pressure	p ₂ [bar] ²⁾	≤ 25

Design details

Design

- Multistage centrifugal pump in ring-section design
- Long-coupled (baseplate-mounted) design for horizontal installation
- Vertical installation with drive lantern or universal-joint shaft

Pump casing

- Suction casing: axial or radial
- Radial suction casing and discharge casing: nozzles can be turned in steps of 90°
- Flanges to EN and ASME (holes and flange facing)
- Stage casings, discharge casings and seal housing sealed by confined O-rings
- Seal cover for mechanical seal

Drive

- 50 Hz and 60 Hz electric motor

Impeller type

- Closed radial impeller with multiply curved vanes

Bearings

- Fixed bearing, drive end, rolling element bearing
- Radial bearing, non-drive end, either plain bearing or rolling element bearing, depending on the type of installation
- Rolling element bearing grease or oil lubricated
- Plain bearing lubricated by fluid handled
- Self-aligning

Coupling

- Long-coupled design, flexible coupling with or without spacer

Coupling guard

Standard:

- Coupling guard, cylindrical

Optional:

- Coupling guard, tread-proof

Shaft seal

- Standardised mechanical seal to EN 12756

Designation

Example: MultiTec Plus³⁾ A 150/5-12.3 18.167 (SP)

Table 2: Designation key

Code	Description
MultiTec Plus	MultiTec Plus type series
A	Installation type
150	Nominal discharge nozzle diameter [mm]
5	Number of stages
12.3	Hydraulic system
18	Material code
167	Seal code
SP	Code for special variants (optional)

1 Do not exceed the maximum solids content of 50 mg/l.

2 The sum of inlet pressure and shut-off head must not exceed the value indicated.

3 Code MTC+

Materials

- Casing: nodular cast iron
- Impellers: duplex steel
- Diffusers: stainless steel

Technical description

Table 3: Installation options A / B / C / D / V

Description	Installation type				
	A	B	C	D	V
	Horizontal design, long-coupled (baseplate-mounted)				Vertical close-coupled pump
	1 shaft passage (drive end)		2 shaft passages		1 shaft passage (drive end)
	Rolling element bearing at the drive end / plain bearing on the suction side		Rolling element bearing at the drive end / rolling element bearing on the suction side	Rolling element bearing at the drive end / rolling element bearing on the discharge side	Rolling element bearing at the drive end / plain bearing on the suction side
	Axial suction nozzle		Radial suction nozzle		
	Drive on discharge side			Drive on suction side	Drive on discharge side
Direction of rotation (seen from the motor end)	Clockwise			Anti-clockwise	Clockwise
Drive	Electric motor				
Axial thrust balancing	By rolling element bearings				
Q_{max} (50 Hz) ⁴⁾	470 m ³ /h				
H_{max}	250 m				
P_{2max}	25 bar				
T_{max}	0 °C to +60 °C				
Shaft seal	Uncooled single mechanical seal				
Casing material	Nodular cast iron				
Hydraulic system material	Diffusers: stainless steel / impellers: duplex stainless steel				

Product benefits

- **Versatile and flexible pump**
 - Variable nozzle positions enable optimal adaptation to site conditions.
- **High operating reliability**
 - A composite wear ring helps reduce internal leakage and extends the service life.
 - The self-aligning plain bearing is made of wear-resistant silicon carbide.
 - Innovative axial thrust balancing supports stable operation.
- **Low operating costs**
 - Optimised hydraulic system and impellers trimmed to the duty point as standard ensure high system efficiency.
 - The KSB SuPremE motor, PumpMeter, and PumpDrive 2 or PumpDrive R variable speed systems ensure optimal efficiency of the complete pump set and automatic adjustment to current demand, enabling significant energy savings.
- **Service-friendly design, low spare parts costs**
 - The bearing and shaft seal can be dismantled easily without removing any hydraulic pump components.
 - Some pump models are designed with a discharge-side seal only, which reduces the costs for purchasing and storing spare parts.
- **Smart monitoring included**
 - All MultiTec Plus pumps are equipped with KSB Guard as standard, enabling continuous condition monitoring and early detection of anomalies.
 - KSB SupremeServ Monitoring Service is included free of charge for the first two years, offering customers valuable insights and proactive service recommendations.
 - The digital service package helps reduce unplanned downtime and optimise maintenance planning.

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

⁴ For 60 Hz values refer to the individual characteristic curves.

Certifications

Table 4: Overview

Label	Effective in:	Comment
	France	Approved in accordance with French drinking water regulations (ACS = Attestation de Conformité Sanitaire)

- Balancing test
- Hydraulic tests:
 - Performance test to ISO 9906 or Hydraulic Institute
 - NPSH test
 - Vibration test
 - Bearing temperature measurement
- Strip test
- Dimensional inspection
- Coating inspection
- Final inspection and testing

Acceptance tests and warranty

Certificates / inspections/ acceptance tests on request:

- Test reports 2.2 to EN 10204 for pressure-retaining components, shaft and impellers
- Inspection certificate 3.1 to EN 10204 for diffusers, impellers and shaft
- Hydrostatic pressure test of pressure-retaining components

Materials

Table 5: Overview of materials

Part No.	Description	Material
106	Suction casing	EN-GJS-400-15
107	Discharge casing	EN-GJS-400-15
108	Stage casing	EN-GJS-400-15
171	Diffuser	1.4408
210	Shaft	1.4021+QT
230	Impeller	1.4517
381/529	Plain bearing	SiC/SiC
502	Casing wear ring	Composite material
523	Shaft sleeve	1.4057+QT
905	Tie bolt	1.7225+QT

Table 6: Materials key

Code / material No.	Description	Standard	To ASTM ⁵⁾
1.4021+QT / X20Cr13+QT	Chrome steel	EN 10088	A276:420
1.4057+QT800 / X17CrNi16-2-QT800	Chrome nickel steel	EN 10088	A276:431
1.4408 / GX5CrNiMo19-11-2	Chrome nickel molybdenum cast steel	EN 10213	A743CF8M
1.4517	Duplex stainless steel	EN 10213	A995 Gr. CD4MCuN
EN-GJS-400-15	Nodular cast iron	EN 1563	A536 Gr. 60_40_18

Seal codes of the mechanical seal

Table 7: Balanced single mechanical seal (uncooled)

Manufacturer	Mechanical seal (code/type)	Material	Static sealing elements	Temperature	Pressure	
					[bar]	
				[°C]	Min.	Max.
KSB	167 (5B)	BQ1EGG	EPDM	≤ 120 °C	-0,5	25
EagleBurgmann	67 (H7N)	Q1BEGG DW001			-0,2	25

Table 8: Key to the material code of the mechanical seal

Code	Material
B	Carbon graphite, resin-impregnated
Q1	Silicon carbide, solid, sintered without pressure
E	EPDM rubber
G	CrNiMo steel

⁵ Similar

Technical data

Table 9: Technical data

Size	Bearings			Diameter of shaft protecting sleeve	Shaft diameter at the coupling	Drive (P/n value)	Other		
	Fixed bearing	Radial bearing	Plain bearing				Hydraulic system	Maximum impeller diameter	Spacer length of spacer-type couplings
				[mm]					
125	3 × 7315	6315 C3	SiC	70	60	0,21	10.3/9.3	262	200
150	3 × 7315	6315 C3	SiC	70	60	0,3	12.3/11.3	325	200

Noise characteristics

Surface sound pressure level LpA: 85 dB, measured at a distance of 1 m from the pump outline (as per DIN 45635 Parts 1 and 24)

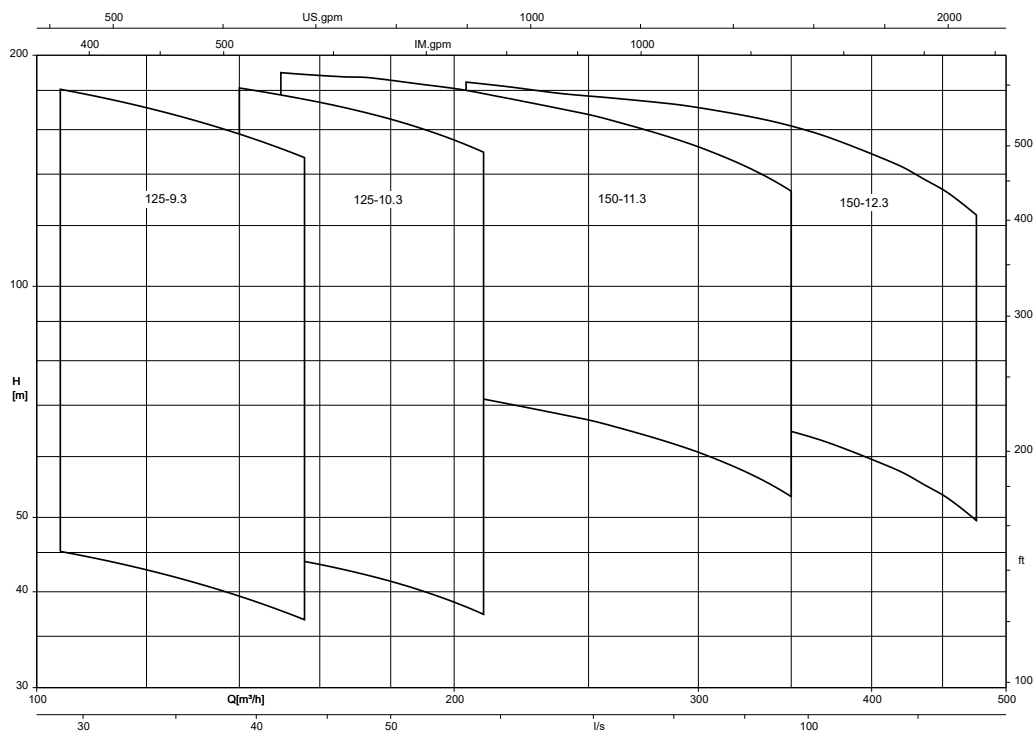
Increase for 60 Hz operation: 3500 rpm +3 dB, 1750 rpm +1 dB

Noise characteristic for other power ratings / speeds: on request

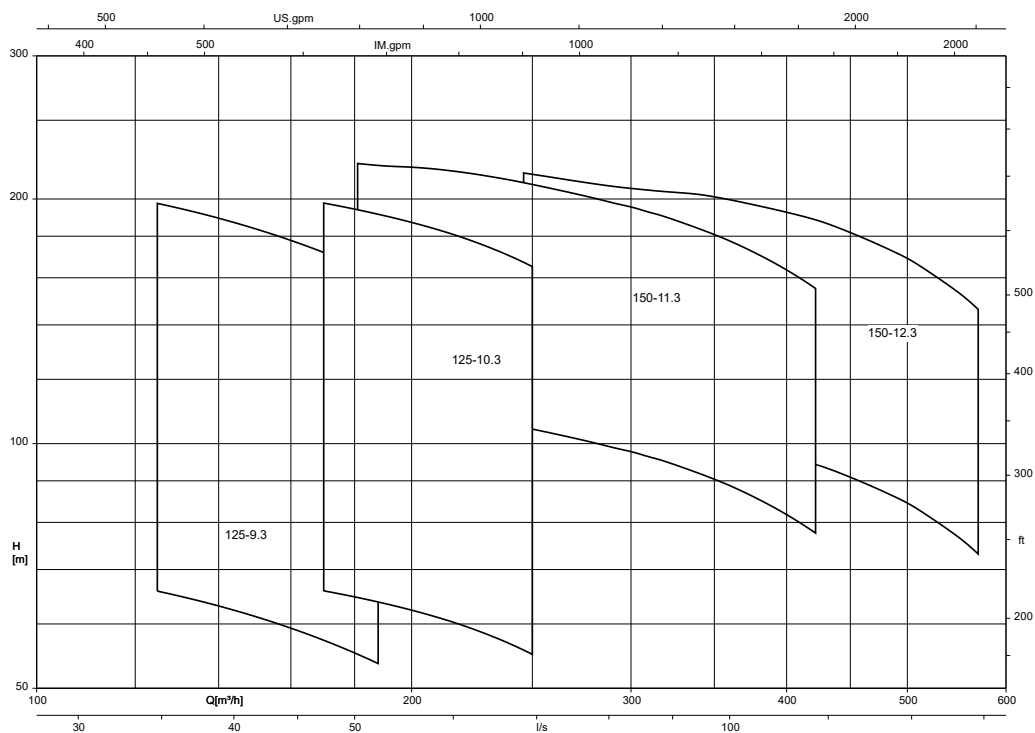
Noise characteristics can only be guaranteed after consultation with the design/engineering department.

Selection charts

MultiTec Plus, 50 Hz, n = 1450 rpm



MultiTec Plus, 60 Hz, n = 1750 rpm



Drive

Table 10: Motor, types of construction

Description	Characteristic
Installation types A / B / C / D	IMB3
Installation type V	V1
Enclosure	IP55 / IP23
Thermal class	F
Pump direction of rotation	Installation types A / B / C / V: clockwise, seen from the drive end
	Installation type D: counter-clockwise, seen from the drive end

Dimensions

Dimensions and connections of MultiTec Plus A / B / C / D

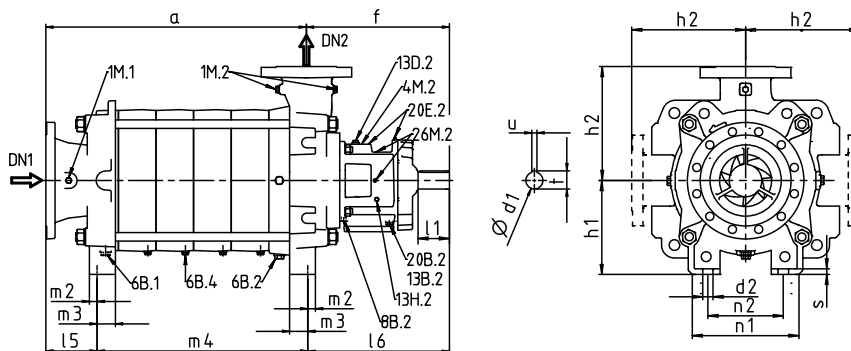


Fig. 1: MultiTec Plus A dimensions

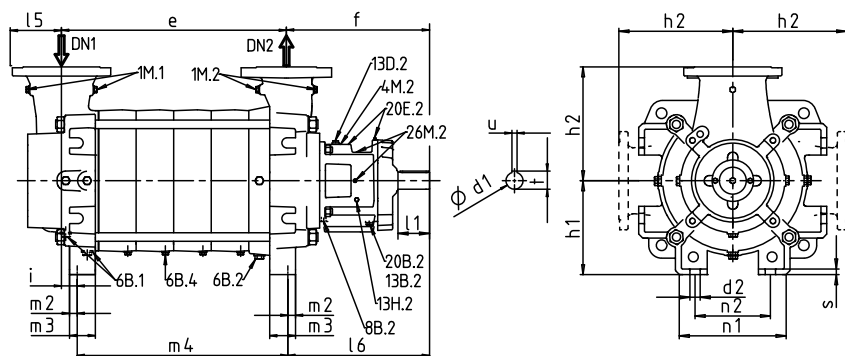


Fig. 2: MultiTec Plus B dimensions

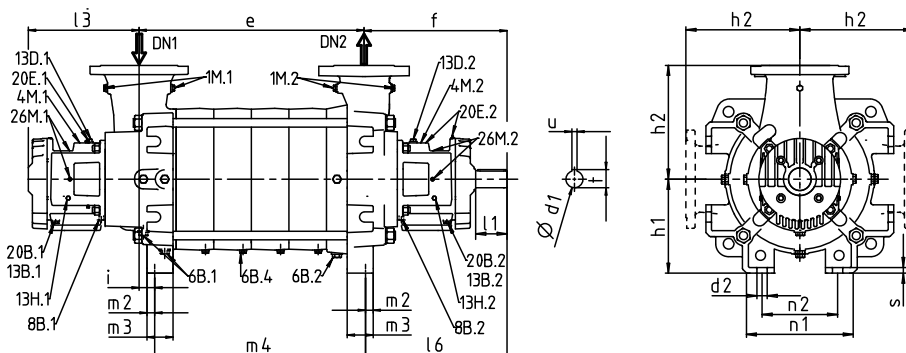


Fig. 3: MultiTec Plus C dimensions

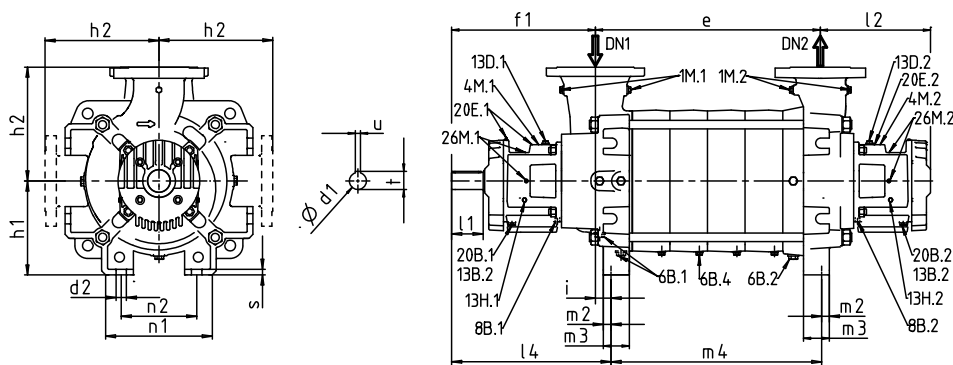


Fig. 4: MultiTec Plus D dimensions

Table 11: Shaft dimensions [mm]

MultiTec Plus A / B / C / D	d _{1k7}	t	u
125	60	64	18
150	60	64	18

Table 12: Dimensions of MultiTec Plus A / B / C / D [mm]

MultiTec Plus A / B / C / D	Number of stages	DN ₁		DN ₂	a	d _{1k7}	d ₂	e	f	f ₁	h ₁	h ₂	i	l ₁	l ₂	l ₃	l ₄	l ₅	l ₆	m ₂	m ₃	m ₄	n ₁	n ₂	s
		Axial	Radial																						
125	2	200	150	125	425	60	24	333	485	518	300	325	44	110	371	403	562	136	472	37	90	303	353	265	20
	3	200	150	125	535	60	24	443	485	518	300	325	44	110	371	403	562	136	472	37	90	413	353	265	20
	4	200	150	125	645	60	24	553	485	518	300	325	44	110	371	403	562	136	472	37	90	523	353	265	20
	5	200	150	125	755	60	24	663	485	518	300	325	44	110	371	403	562	136	472	37	90	633	353	265	20
	6	200	150	125	865	60	24	773	485	518	300	325	44	110	371	403	562	136	472	37	90	743	353	265	20
	7	200	150	125	975	60	24	883	485	518	300	325	44	110	371	403	562	136	472	37	90	853	353	265	20
	8	200	150	125	1085	60	24	993	485	518	300	325	44	110	371	403	562	136	472	37	90	963	353	265	20
	2	250	200	150	519	60	36	394	502.5	505	330	400	55	110	387	389	560	180	498	27	91	344	375	265	20
150	3	250	200	150	651	60	36	526	502.5	505	330	400	55	110	387	389	560	180	498	27	91	476	375	265	20
	4	250	200	150	783	60	36	658	502.5	505	330	400	55	110	387	389	560	180	498	27	91	608	375	265	20
	5	250	200	150	915	60	36	790	502.5	505	330	400	55	110	387	389	560	180	498	27	91	740	375	265	20

Table 13: Pressure gauge, vibration and temperature measurement connections

Connection ⁶⁾	MultiTec Plus A / B / C / D
	125 / 150
1M.1	G 1/2 ⁷⁾
1M.2	G 1/2 ⁷⁾
26M.1 & 2	M8x125
4M.1 & 2	G 1/4

Table 14: Drain connections

Connection ⁶⁾	MultiTec Plus A / B / C / D	
	125	150
6B.1	G 1/2 ⁷⁾	G 1 ⁷⁾
6B.2	G 1/2 ⁷⁾	G 1 ⁷⁾
6B.4	G 1/2 ⁷⁾	G 1/2 ⁷⁾
8B.1 & 2	G 3/8	G 3/8

Table 15: Lubrication connections

Connection ⁶⁾		MultiTec Plus A / B / C / D	
		125 / 150	
		Grease	Oil
20E.1 & 2	M8x100	Lubricating nipple	Plugged
13D.1 & 2	G 1/2	Plugged	Vent plug
20B / 13B.1 & 2	G 1/2	Plugged	Plugged
13H.1 & 2	G 1/4	Plugged	Constant level oiler

6 G = ISO 228/1

7 Plugged

MultiTec Plus V dimensions

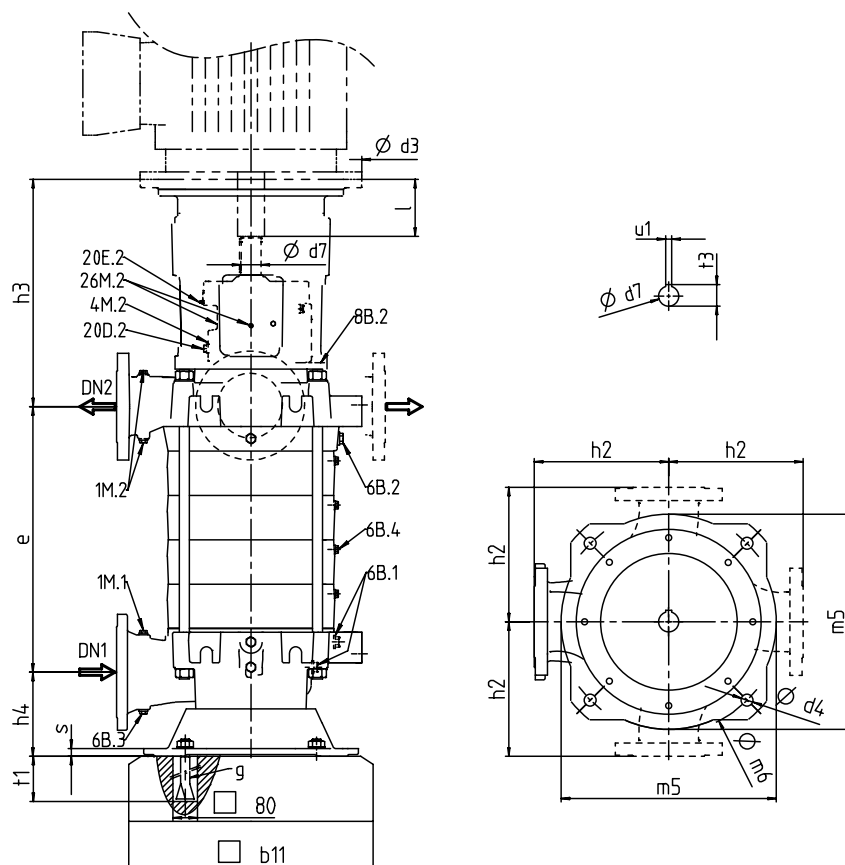


Fig. 5: MultiTec Plus V dimensions

Table 16: Shaft dimensions [mm]

MultiTec Plus V	d_{7k7}	t_3	u_1
125	60	64	18
150	60	64	18

Table 17: MultiTec Plus V dimensions [mm]

MultiTec Plus V	Number of stages	DN ₁	DN ₂	b ₁₁	d ₄	e	g	h ₂	h ₄	m ₅	m ₆	s	t ₁
125	2	150	125	800	4xØ36x90°	333	M30×400 MU	325	264	640	Ø660	23	400
	3	150	150	800	4xØ36x90°	443	M30×400 MU	325	264	640	Ø660	23	400
	4	150	150	800	4xØ36x90°	553	M30×400 MU	325	264	640	Ø660	23	400
	5	150	150	800	4xØ36x90°	663	M30×400 MU	325	264	640	Ø660	23	400
	6	150	150	800	4xØ36x90°	773	M30×400 MU	325	264	640	Ø660	23	400
	7	150	150	800	4xØ36x90°	883	M30×400 MU	325	264	640	Ø660	23	400
	8	150	150	800	4xØ36x90°	993	M30×400 MU	325	264	640	Ø660	23	400
150	2	200	150	800	4xØ36x90°	394	M30×400 MU	400	250	640	Ø660	23	400
	3	200	150	800	4xØ36x90°	526	M30×400 MU	400	250	640	Ø660	23	400
	4	200	150	800	4xØ36x90°	658	M30×400 MU	400	250	640	Ø660	23	400
	5	200	150	800	4xØ36x90°	790	M30×400 MU	400	250	640	Ø660	23	400

Table 18: Pressure gauge, vibration and temperature measurement connections

Connection ⁸⁾	MultiTec Plus V
	125 / 150
1M.1	G 1/2 ⁹⁾
1M.2	G 1/2 ⁹⁾
26M.2	M8x125
4M.2	G 1/4

Table 19: Drain connections

Connection ⁸⁾	MultiTec Plus V	
	125	150
6B.1	G 1/2 ⁹⁾	G 1 ⁹⁾
6B.2	G 1/2 ⁹⁾	G 1 ⁹⁾
6B.3	G 1/2 ⁹⁾	G 1/2 ⁹⁾
6B.4	G 1/2 ⁹⁾	G 1/2 ⁹⁾
8B.2	G 3/8	G 3/8

Table 20: Lubrication connections

Connection ⁸⁾		MultiTec Plus V
		125 / 150
		Grease
20E.2	M8x100	Lubricating nipple
20B.2	G 1/2	Plugged

Variable dimensions for motors

Table 21: Variable dimensions for motors

Motor			
d ₃	l	h ₃	
		125	150
F740	170	661	677,5
F600	170	660	676,5
F500	140	630	646,5
F400	140	630	646,5
F350	110	630	-

Flange design (standard)

Flange castings not in accordance with EN or ASME. Only seal face and bolt hole pattern according to standard.

Table 22: Flange designs (standard)

Material variant	Flanges drilled to EN			Flanges drilled to ASME Class		
	Standard	Suction flange	Discharge flange	Standard	Suction flange	Discharge flange
		PN	PN			
18	EN 1092-2	16	25	ASME B 16.1	125 RF	250 RF

8 G = ISO 228/1

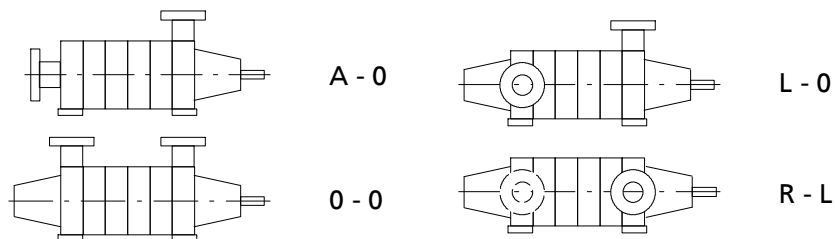
9 Plugged

Nozzle position

The nozzle position is variable. The required nozzle position has to be selected when completing an order in the selection program.
Nozzle position 0-0 (or illustration 2 for vertical installation) is available for all pump sizes and material variants.

Horizontal installation (A, B, C, and D)

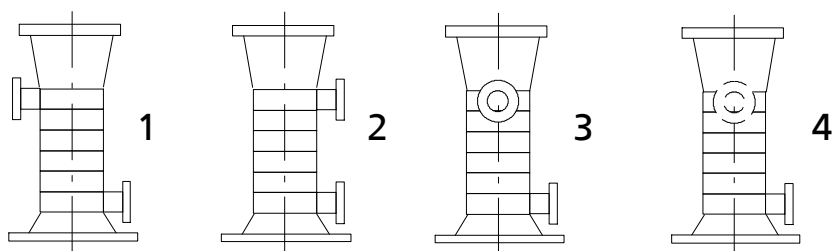
The first letter defines the suction nozzle, the second letter the discharge nozzle.



A	Axial suction nozzle
0	Suction nozzle and/or discharge nozzle on top
R	Suction nozzle and/or discharge nozzle on the right
L	Suction nozzle and/or discharge nozzle on the left

Vertical installation

The suction nozzle (at the bottom) serves as a fixed point. The illustration number indicates the offset position of the discharge nozzle compared to the suction nozzle.



1	Offset by 180°
2	Aligned in same direction
3	90° offset to the left
4	90° offset to the right

General assembly drawing with list of components (example)

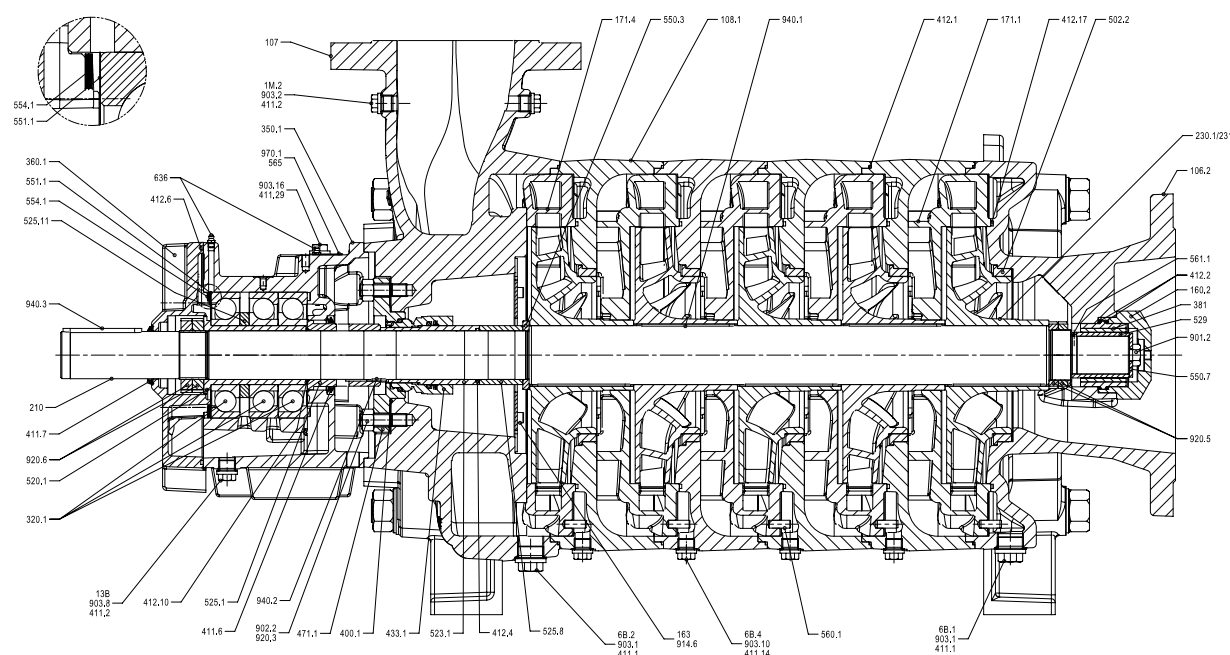


Fig. 6: General assembly drawing

Table 23: List of components

Part No.	Description	Part No.	Description
106.2	Suction casing	523.1	Shaft sleeve
107	Discharge casing	525.1	Spacer sleeve
108.2	Stage casing	529	Bearing sleeve SiC
160.2	Cover	540.1	Bush
171.1	Diffuser	550.3/.4/.7	Disc
171.4	Diffuser, last stage	551.1	Spacer disc
210	Shaft	554.1	Spring washer
230.1	Impeller	560.1/.2	Grooved pin
320.1	Rolling element bearing	565	Rivet
350.1	Bearing housing	636	Lubricating nipple
360.1	Bearing cover	81-92	Cover plate
381	Bearing cartridge	900.4	Screw
400.1	Gasket	901.1/.2	Hexagon head bolt
411.1/.2/.3/.4/.7	Joint ring	902.1/.2	Stud
412.1/.2/.4/.10	O-ring	903.1/.2	Screw plug
433.1	Mechanical seal	905	Tie bolt
471.1	Seal cover	914.1	Hexagon socket head cap screw
502.1	Casing wear ring	920.1/.3/.4/.5	Nut
507	Thrower	940.1/.2/.3	Key
520.1	Sleeve	970.1	Label/plate



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